

Empirical Evaluation of ProtoHedge: Interpretable Deep Hedging with Real Market Data

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ABSTRACT

ProtoHedge was introduced as an interpretable alternative to black-box Deep Hedging, but its original evidence was confined to synthetic market environments. This paper studies whether prototype-based hedging remains competitive and interpretable on historical option-hedging episodes. We construct a real-data panel of 30,797 20-step episodes across ten U.S. equity and ETF underlyings, and compare ProtoHedge with unhedged exposure, delta-style rules, and a matched Deep Hedging benchmark under transaction costs, step-wise accounting, hard trading bounds, validation-based model selection, and three random seeds. The mean-oriented ProtoHedge frontier has the strongest average mean gains in the completed European-call panel, while the tail-oriented frontier has the strongest empirical CVaR_{5%}. Interpretability is verified directly through prototype concentration, regime-dependent usage, and nearest historical episodes. We also specify the path-aware Asian-call extension required to test these conclusions for a path-dependent liability; the incomplete preliminary Asian run is not used for the paper’s main claims.

CCS CONCEPTS

• **Computing methodologies** → **Instance-based learning**; *Sequential decision making*; • **Applied computing** → *Economics*.

KEYWORDS

deep hedging, interpretable machine learning, prototype learning, option hedging, empirical finance

1 INTRODUCTION

Deep hedging learns a sequential trading policy directly from market states, trading frictions, and a risk-sensitive objective. This flexibility is useful when hedging is discrete, costly, and constrained, but a flexible neural policy can be difficult to inspect. ProtoHedge addresses that problem by routing hedge decisions through a finite set of representative market states, or prototypes. The original ProtoHedge evidence was synthetic; the central question in this paper is whether its performance–interpretability tradeoff survives on real historical option data.

Our contribution is empirical rather than architectural. We evaluate the fixed-prototype design under explicit transaction costs, step-wise accounting, trade and cumulative-position bounds, validation-based checkpoint selection, and multi-seed testing. The goal is to determine whether ProtoHedge remains competitive with a matched black-box Deep Hedging benchmark and whether its prototypes continue to correspond to inspectable market states.

2 PROTOHEDGE AND THE EMPIRICAL VALIDATION PROBLEM

This section places our study in context. We first summarize the deep hedging framework and the role of vanilla Deep Hedging as the main black-box benchmark. We then explain the ProtoHedge architecture and its interpretability motivation. Finally, we identify the central gap left by the original ProtoHedge study: although prototype-based hedging is attractive conceptually, it must still be shown to remain competitive and interpretable on real historical market data.

2.1 Deep Hedging as a Learning Problem

Deep hedging treats hedging as a sequential decision problem rather than a closed-form replication problem. At each trading time t , the hedger observes a state vector

$$x_t \in \mathcal{X},$$

which may contain the underlying price, option sensitivities, implied volatility, time-to-maturity information, current hedge inventory, and other market features. A policy

$$\pi_\theta : \mathcal{X} \rightarrow \mathcal{A}$$

with parameters θ maps the observed state to a hedge action a_t , for example a change in the position held in the underlying or in liquid hedging instruments. Over an episode of length T , these actions generate a cumulative hedging profit and loss after transaction costs and constraints. The learning problem is to choose θ so as to optimize a risk-sensitive objective on terminal hedging outcomes.

In modern deep hedging, this policy is typically represented by a neural network and trained end-to-end. This approach is attractive because it does not require the world to be exactly Black–Scholes, does not assume frictionless continuous rebalancing, and can incorporate practical features such as transaction costs, trading bounds, incomplete hedge sets, and high-dimensional state variables. In this sense, deep hedging generalizes classical delta hedging: instead of prescribing a rule analytically from a stylized model, it learns a rule directly from data and the training objective.

Let G^π denote the total hedging gain of a policy π over an episode, net of liability payoff and trading costs. A standard deep hedging problem can be written abstractly as

$$\max_{\theta} \mathbb{E}[U(G^{\pi_\theta})] \quad \text{or} \quad \min_{\theta} \rho(-G^{\pi_\theta}),$$

where U is a utility function or ρ is a risk measure such as CVaR. In practice, the exact parameterization varies across implementations, but the central idea is the same: learn a hedging policy that performs well under realistic frictions and constraints, rather than relying on an idealized replication formula.

The principal weakness of this framework is interpretability. A sufficiently expressive neural policy can produce strong hedging performance while remaining difficult to inspect. Even when one can evaluate its actions numerically, it may still be unclear which market situations drive those actions, whether similar states are treated similarly, and whether the policy has learned economically meaningful behavior or merely a complicated nonlinear mapping that is hard to diagnose. This matters especially in hedging, where large position changes, boundary-seeking behavior, and sensitivity to tail events are economically significant. For that reason, vanilla Deep Hedging is a natural and important benchmark, but it is not fully satisfactory when interpretability is part of the modeling objective.

2.2 ProtoHedge: Prototype-Based Deep Hedging

ProtoHedge was introduced as an interpretable alternative to a fully black-box deep hedging policy. Its key idea is to force the policy to make decisions through comparisons with a small set of representative states, called prototypes, rather than through an unrestricted latent representation alone. This changes the nature of the model explanation: instead of saying only that a neural network took a certain action, one can ask which stored prototype states the current market state most closely resembles, and how those prototypes influence the hedge.

At a high level, ProtoHedge still solves the same sequential hedging problem as vanilla Deep Hedging. The state x_t is observed at time t , the policy outputs an action a_t , and the model is trained under a risk-sensitive objective. The difference lies in the internal architecture. Rather than feeding the state directly through a generic multilayer network to produce an action, ProtoHedge first compares the current state to a collection of prototypes

$$\{p_1, \dots, p_K\},$$

where each p_k is a representative point in feature space. These comparisons produce similarity scores

$$s_k(x_t),$$

which are then aggregated to determine the action. In simplified form, the action can be viewed as a prototype-weighted combination

$$a_t \approx \sum_{k=1}^K w_k(x_t) \alpha_k,$$

where $w_k(x_t)$ reflects the similarity between the current state and prototype p_k , and α_k is the prototype-associated hedge response. The exact architecture can include additional nonlinear layers and weighting choices, but this representation captures the central intuition: the hedge is routed through a small library of representative market states.

This design is important for two reasons. First, it creates an explicit bottleneck in the policy. If only a few prototypes dominate the decision at a given state, then the model's behavior becomes more inspectable. Second, the prototypes themselves can be examined. One can ask what kind of market situations they represent, how frequently they are used, whether they specialize by regime, and whether they correspond to recognizable economic conditions. In

contrast to a generic hidden layer, the prototype layer is intended to admit example-based interpretation.

The original ProtoHedge work demonstrated that this structure can preserve much of the performance of vanilla Deep Hedging in synthetic settings while providing a more interpretable decision mechanism. Conceptually, ProtoHedge is appealing because it does not reject deep hedging; instead, it restructures it. The goal is not to eliminate expressiveness, but to trade some unrestricted flexibility for a representation that is easier to inspect and potentially more economically meaningful.

This distinction is critical for our paper. We are not testing whether prototype-based hedging is a fundamentally different problem from deep hedging. We are testing whether the specific interpretability mechanism introduced by ProtoHedge remains viable once one moves beyond controlled simulated worlds.

2.3 Why Empirical Validation Is the Missing Step

The original ProtoHedge results are important, but they do not by themselves establish that the method is useful in realistic hedging environments. Synthetic experiments are valuable because they allow careful control over the data-generating process and clean benchmarking against known baselines. However, they do not fully capture the empirical conditions under which hedging systems are actually deployed.

Historical market data introduce several sources of difficulty that do not appear cleanly in stylized simulation. Market dynamics are non-stationary; realized option paths reflect changing volatility regimes, crisis periods, and structural breaks; transaction costs and rebalancing constraints matter; and the available hedge signals are noisy and imperfect. Moreover, the interpretability question becomes more demanding on real data. In a synthetic environment, a prototype may be interpretable because the world itself is low-dimensional and model-generated. In historical data, a prototype is only useful if it corresponds to a recognizable market state and if its usage is stable enough to support economic interpretation.

This creates three empirical questions that the original ProtoHedge paper leaves open.

First, does ProtoHedge remain competitive with vanilla Deep Hedging on held-out historical data? A prototype bottleneck is only worthwhile if it preserves most of the risk-management quality of the black-box benchmark.

Second, does the prototype structure behave sensibly under realistic constraints? A learned hedging policy can appear strong in aggregate while relying excessively on hard position limits or other boundary effects. In the empirical setting, such behavior is economically important, because a model that achieves performance only by living at its constraints is less credible than one that performs similarly in the interior of the feasible region.

Third, does the interpretability claim survive contact with real data? If ProtoHedge is to be more than a proof-of-concept, then prototype usage should be concentrated enough to be meaningful, should vary in an intelligible way across market regimes, and should be relatable to actual historical episodes rather than only abstract feature vectors.

For these reasons, empirical validation is not a minor extension but a necessary step in evaluating ProtoHedge as a hedging framework.

2.4 What This Paper Tests

This paper studies ProtoHedge precisely along that missing empirical axis. We evaluate it on real historical hedging episodes constructed from market data, compare it against classical baselines and vanilla Deep Hedging, and examine whether its prototype-based interpretability mechanism remains meaningful outside synthetic settings.

Our empirical design focuses on two liabilities. The primary task is hedging a short at-the-money European call over fixed-length historical episodes. We then extend the same framework to a path-dependent short at-the-money Asian call, which tests whether the conclusions persist when the liability depends on the realized path of the underlying rather than only its terminal value. In both cases, the goal is not merely to report raw performance, but to assess the performance–interpretability tradeoff under realistic frictions and constraints.

More specifically, the paper tests the following claims.

- (1) ProtoHedge remains competitive with vanilla Deep Hedging on real historical data across a panel of underlyings, rather than only in synthetic environments.
- (2) ProtoHedge can improve or preserve downside risk characteristics without relying more heavily than vanilla on hard trading or position boundaries.
- (3) The prototype layer remains empirically interpretable: usage is concentrated in a relatively small subset of prototypes, varies systematically across market regimes, and can be connected to recognizable historical states.
- (4) These conclusions are not limited to a terminal European-call liability, but extend to a path-dependent Asian-call setting under the same evaluation framework.

This framing also clarifies what the paper does *not* attempt to show. Our objective is not to argue that ProtoHedge universally dominates vanilla Deep Hedging on every metric and every market. Nor is our contribution to propose a new prototype architecture. Rather, our contribution is to test whether the central ProtoHedge idea — interpretable prototype-based deep hedging — remains viable and informative in an empirical setting where realistic market frictions, regime changes, and historical path structure matter.

The remainder of the paper operationalizes that question. Section 3 describes the real-data experimental design, including the construction of historical episodes, the liabilities, the model set, and the training protocol. Section 4 then reports the real-data results, including panel-level performance, position-bound behavior, interpretability diagnostics, and the path-dependent Asian-option extension.

3 EXPERIMENTAL FRAMEWORK

This section describes the empirical design used to evaluate ProtoHedge on historical market data. Our implementation is deliberately conservative. We use fixed-length out-of-sample hedging episodes, explicit transaction costs, bounded trading actions, bounded cumulative hedge positions, validation-based model selection, and

multi-seed evaluation. The goal is not merely to train a profitable hedge in sample, but to assess whether ProtoHedge remains competitive with vanilla Deep Hedging under a realistic and reproducible real-data protocol.

3.1 Data and Episode Construction

Our real-data experiments are based on historical U.S. equity and ETF option data obtained through WRDS, combining underlying spot-price histories with daily listed option records. The final panel contains ten liquid underlyings spanning broad equity indices, sector ETFs, fixed income, and single-name equities:

$$\{\text{AAPL, IWM, NVDA, QQQ, SPY, TLT, XLE, XLF, XLK, XLV}\}.$$

For each ticker, the cleaned feature panel runs from 2005-01-11 to 2024-12-31.

The raw option-chain data are filtered to call options and aligned with the corresponding daily spot series. For each date, we select a single liquid near-at-the-money call contract by sorting candidate options according to:

- (1) absolute distance between strike and spot,
- (2) absolute distance between the contract’s days-to-expiration and a 30-day target maturity,
- (3) bid–ask spread,
- (4) open interest (descending), and
- (5) volume (descending),

subject to a maturity filter of 20 to 40 calendar days to expiration. This yields a daily panel of one representative short-dated near-ATM call per underlying.

The resulting per-date feature vector is

$$x_t^{\text{raw}} = (S_t, C_t, \Delta_t^{\text{call}}, v_t, \sigma_t^{\text{impl}}),$$

where S_t is the underlying spot, C_t is the selected call price, Δ_t^{call} is the selected call delta, v_t is the selected call vega, and σ_t^{impl} is implied volatility. These features are stored in the episode tensor in the fixed order

$$(\text{spot}, \text{call_price}, \text{call_delta}, \text{call_vega}, \text{ivol}).$$

Historical hedging episodes are constructed as rolling 20-step windows. Importantly, windows are created only within contiguous trading-date blocks: whenever the gap between consecutive observation dates exceeds four calendar days, a new block is started. This prevents the episode builder from stitching together disjoint option cycles separated by large holes in the cleaned panel. Letting $\tau = 20$ denote the episode length, each final sample is a tensor in

$$\mathbb{R}^{\tau \times 5}.$$

Across the ten-ticker panel, this procedure yields 30,797 usable 20-step episodes in total.

Although the raw episode tensor contains five market features, the learning models do not consume all five directly. The hedging agents are trained on the state representation used by the deep hedging gym, namely current hedgeable prices, current hedge inventory, and time remaining. Concretely, the learned policies use the feature list

$$(\text{price}, \text{delta}, \text{time_left}),$$

where price is the two-dimensional vector of current spot and current selected call price, delta denotes the current cumulative hedge position in the two traded instruments, and time_left is the remaining time to maturity. The richer market panel remains important because it defines the liability, the classical baselines, the transaction-cost model, and the regime-analysis variables.

Finally, for numerical stability and cross-path comparability, the real-data world normalizes spot, call price, and call vega by each path's initial spot level. Under this normalization, a liability struck at-the-money at inception has normalized strike one, while remaining equivalent to a strike of S_0 in raw price units.

3.2 Hedging Tasks

We study two liabilities on the same historical episode panel.

European ATM short call. The primary task is to hedge a short at-the-money European call over a 20-step historical episode. If $S_{i,0}$ denotes the initial spot on episode i , then the strike is set to

$$K_i = S_{i,0}.$$

The liability payoff at maturity is

$$H_i^{\text{Euro}} = -\max(S_{i,T} - K_i, 0),$$

where the negative sign reflects the fact that the hedger is short the option.

Asian ATM short call. The path-dependent extension replaces the European liability with an arithmetic-average Asian call, again struck at-the-money at episode start:

$$K_i = S_{i,0}.$$

Let the episode contain T trading decisions. The Asian liability is

$$H_i^{\text{Asian}} = -\max\left(\frac{1}{T} \sum_{t=1}^T S_{i,t} - K_i, 0\right).$$

In the implementation used for the final paper sweep, the averaging window spans the full historical episode. Because the payoff depends on the accumulated path, a fully observed Asian-policy state must also include the causal running average

$$\bar{S}_{i,t}/K_i = \frac{1}{t+1} \sum_{u=0}^t S_{i,u}/K_i.$$

This quantity is observable at time t and does not leak future information. It must be supplied to both Deep Hedging and ProtoHedge so that their comparison remains fair.

Hedge instruments and actions. The hedge may trade in two liquid instruments:

- (1) the underlying spot instrument, and
- (2) the selected near-ATM listed call option.

The action at each step is therefore two-dimensional,

$$a_t = (a_t^{(S)}, a_t^{(C)}),$$

where each component denotes a trade increment rather than an absolute target position. The cumulative hedge inventory evolves recursively as

$$\delta_t = \delta_{t-1} + a_t.$$

Step-wise hedge accounting. All reported experiments use step-wise hedge accounting rather than a terminal shortcut. If ΔS_t and ΔC_t denote the next-step changes in the spot and option hedge instruments, then the pathwise hedge P&L is accumulated as

$$\text{PnL}_i = \sum_{t=1}^T \left(a_{i,t}^{(S)} \Delta S_{i,t} + a_{i,t}^{(C)} \Delta C_{i,t} \right).$$

This choice is important empirically. In earlier exploratory real-data runs, more permissive accounting and looser constraints could produce apparently strong results together with unrealistic cumulative-bound saturation. The current step-wise setup is intentionally stricter and better aligned with the actual sequence of hedge rebalancing decisions.

Transaction costs. Trading costs are imposed separately on spot and option trades. The per-unit spot trading-cost coefficient is

$$c_{i,t}^{(S)} = 0.0002 |S_{i,t}|.$$

The per-unit option trading-cost coefficient is

$$c_{i,t}^{(C)} = 0.02 |v_{i,t}| + 0.0002 |\Delta_{i,t}^{\text{call}}| + 0.0005 |C_{i,t}|.$$

The realized trading cost on episode i is therefore

$$\text{Cost}_i = \sum_{t=1}^T \left(|a_{i,t}^{(S)}| c_{i,t}^{(S)} + |a_{i,t}^{(C)}| c_{i,t}^{(C)} \right).$$

Gains. For both liabilities, the terminal hedging gain is defined as

$$G_i = H_i + \text{PnL}_i - \text{Cost}_i.$$

This is the central outcome used throughout the empirical evaluation.

3.3 Models and Baselines

We compare ProtoHedge against both classical baselines and a black-box neural benchmark.

Unhedged. This baseline takes no hedge action:

$$a_t \equiv 0.$$

Its gain is simply the liability payoff.

Spot-Delta. This is a rule-based baseline that hedges only with the underlying spot instrument. At each step, the target cumulative spot position is set equal to the observed delta of the selected near-ATM call. The option hedge position remains zero. This baseline uses the market-provided option delta directly and therefore provides a transparent classical reference point.

Spot-Delta Band. This baseline extends Spot-Delta with a no-trade band. Instead of rebalancing to the point target at every step, the current cumulative spot position is left unchanged as long as it remains within

$$[\Delta_t^{\text{call}} - b, \Delta_t^{\text{call}} + b],$$

where $b \geq 0$ is a band width. If the current position exits the band, it is projected back to the closest boundary. We tune b on the validation set using the grid

$$b \in \{0.00, 0.01, 0.02, 0.05, 0.10, 0.15\},$$

selecting the band that maximizes validation mean gains.

Vanilla Deep Hedging. The main black-box benchmark is a feed-forward neural hedging policy trained end-to-end under the same world, friction model, and objective as ProtoHedge. The vanilla agent uses the state representation

$$(\text{price}, \text{delta}, \text{time_left}),$$

and outputs a two-dimensional trade increment. In the reported panel runs, the vanilla network has width 20, depth 3, and Softplus activations.

ProtoHedge. ProtoHedge uses the same state representation and output space as vanilla Deep Hedging, but constrains action generation through a prototype layer. Each current state is compared to a finite collection of stored prototypes, and the resulting similarity pattern determines the hedge action. In the reported full grid, we vary:

- (1) the number of prototypes

$$K \in \{10, 25, 50, 100\},$$
- (2) the prototype source policy, using either a trained vanilla hedge or the Spot-Delta baseline as the source from which prototypes are extracted, and
- (3) whether similarity is weighted or unweighted.

Prototype extraction follows the same reusable pipeline throughout the repo. For a given source policy, we:

- (1) construct the flattened feature matrix seen by the policy,
- (2) standardize it with a StandardScaler,
- (3) apply K -means clustering in standardized feature space, and
- (4) store, for each cluster, the nearest real standardized feature vector rather than the raw cluster centroid.

This last step matters because it keeps each stored prototype tied to an actually observed market state rather than to an abstract average point with no direct empirical counterpart.

In the reported real-data panel runs, learned distance-feature weights are held fixed, i.e. we do not yet let prototypes or feature-weight vectors move during training. This keeps the present paper focused on empirical validation of the original fixed-prototype design.

3.4 Training and Model Selection Protocol

All real-data neural models are trained under the same multi-seed protocol.

Train/validation/test splits. For each ticker and each random seed, the episode tensor is randomly permuted at the path level and split into

$$70\% \text{ train}, \quad 15\% \text{ validation}, \quad 15\% \text{ test}.$$

The main reported panel uses the three seeds

$$(1234, 2345, 3456).$$

Optimization. All neural models are trained for 800 epochs using the same CVaR-based deep hedging objective with risk-aversion parameter one. The optimizer uses learning rate

$$10^{-3},$$

together with clipping safeguards already built into the training pipeline. Classical baselines are not trained in the neural sense, but

they are evaluated on exactly the same splits and under the same friction model.

Feature normalization and state semantics. The real-data world is run in normalized mode. Spot, call price, and call vega are divided pathwise by the initial spot, while the state passed to the learning agent at each step consists of current hedgeable prices, current hedge inventory, and time remaining. This keeps the neural state compact and comparable across underlyings while preserving the original market information needed for payoff definition, baseline construction, and regime analysis.

Validation-based checkpoint selection. Model selection is based on the validation split, not on the training loss alone. Let L_{val} denote the validation objective. For the final robust panel runs, checkpoints are ranked by the selection score

$$S = L_{\text{val}} + \alpha_a \overline{|a|} + \alpha_\delta \overline{|\delta|} + \alpha_b \text{BO} + \alpha_p \text{PT},$$

where $\overline{|a|}$ is mean absolute action magnitude on the validation set, $\overline{|\delta|}$ is mean absolute cumulative position magnitude, BO is validation bound occupancy, and PT is the fraction of validation paths that touch any cumulative bound. In the reported full runs, we use

$$\alpha_a = 0.005, \quad \alpha_\delta = 0.010, \quad \alpha_b = 0.100, \quad \alpha_p = 0.$$

Thus, validation selection explicitly favors models that do not rely excessively on large actions, large cumulative positions, or persistent boundary saturation.

Regularization. In addition to validation-time selection penalties, the training loss includes mild regularization terms on action and cumulative-position magnitude:

$$\lambda_a = 0.001, \quad \lambda_\delta = 0.002.$$

Concretely, if L_{hedge} is the base objective, then the optimized loss is

$$L_{\text{train}} = L_{\text{hedge}} + \lambda_a \mathbb{E}[|a|] + \lambda_\delta \mathbb{E}[|\delta|].$$

These penalties are deliberately mild: they do not dominate the objective, but they discourage the model from achieving performance through unnecessarily aggressive trading.

Trade bounds and cumulative position bounds. The final reported real-data runs impose both action-level and cumulative-position bounds. At every step and for each traded instrument, trade increments are clipped to

$$[-1, 1],$$

and cumulative hedge positions are also constrained to

$$[-1, 1].$$

These bounds are scientifically important. In exploratory unconstrained runs, apparent performance improvements could be accompanied by heavy saturation at the position boundaries. The bounded protocol is designed to test whether ProtoHedge remains competitive without relying on unrealistic edge-of-feasible-region behavior.

Paper sweep design. For the European-call panel, the final full sweep evaluates the full ProtoHedge grid described above across all ten tickers and all three seeds. The completed Asian-call extension will use the same grid and seed schedule, with the causal running-average feature added to both learned policies. This keeps the European and Asian comparisons identical except for the information required by the path-dependent liability.

3.5 Evaluation Metrics

We evaluate models at both the hedging-performance level and the interpretability level.

Primary performance metrics. For each fitted model and each test split, we compute:

- (1) mean gains,

$$\mathbb{E}[G],$$
- (2) the empirical 5% Value-at-Risk and Conditional Value-at-Risk of gains,

$$\text{VaR}_{5\%}(G), \quad \text{CVaR}_{5\%}(G),$$
- (3) shortfall probability,

$$\mathbb{P}(G < 0),$$
- and
- (4) realized mean trading-cost and hedge-P&L summaries.

In the paper narrative, mean gains and $\text{CVaR}_{5\%}$ are the headline performance measures. Shortfall probability is retained as a secondary summary rather than the sole criterion of success.

Position-bound diagnostics. Because constrained hedging behavior is a central part of the empirical question, we also record:

- (1) the fraction of all path-step-instrument positions lying at either cumulative position bound,
- (2) the fraction lying specifically at the upper or lower bound, and
- (3) the fraction of paths that touch any cumulative position bound at least once.

We refer to the first quantity as *bound occupancy*. Lower bound occupancy is interpreted as evidence that a model is operating more consistently in the interior of the feasible region, which is economically preferable all else equal.

Robust-screen criterion. When constructing paper frontiers among ProtoHedge models, we apply a robustness screen based on average bound occupancy and path-touch behavior. In the final reported panel setup, models with excessive bound saturation are excluded from the empirical frontier even if they perform well on raw mean gains. This prevents the final comparison from being driven by unrealistic edge-seeking solutions.

Prototype concentration metrics. For ProtoHedge models, we further compute:

- (1) average prototype weight across all paths and time steps,
- (2) top-1 prototype frequency, i.e. the proportion of decisions for which each prototype receives the highest weight, and
- (3) cumulative usage mass of the top- k prototypes.

These metrics quantify whether decisions are routed through a small number of dominant prototype states or diffusely across many prototypes.

Regime-sensitive interpretability metrics. To test whether prototype use changes systematically with market conditions, we form episode-level regime labels from the historical paths:

- *return regime*, using terciles of terminal return,
- *volatility regime*, using terciles of average implied volatility, and
- *drawdown regime*, using terciles of maximum realized drawdown.

The resulting labels are *down*, *middle*, and *up* for returns; *low*, *mid*, and *high* volatility; and *mild*, *mid*, and *deep* drawdown. Within each regime, we summarize gains, tail risk, shortfall, and average prototype weights. This lets us test whether the same few prototypes dominate everywhere, or whether ProtoHedge reallocates attention across market states in a regime-sensitive way.

Nearest historical states. Finally, for the dominant prototypes, we recover the nearest historical feature vectors and corresponding underlying paths. This allows us to move beyond abstract prototype coordinates and ask whether the most-used prototypes correspond to recognizable empirical market scenarios. This step is central to the interpretability claim of the paper: the prototype mechanism is not merely sparse, but empirically inspectable.

4 REAL-DATA EMPIRICAL RESULTS

This section evaluates whether ProtoHedge preserves a favorable performance-interpretability tradeoff on historical option-hedging episodes. All reported European-option results use the common protocol described in Section 3: three random seeds, held-out test episodes, step-wise hedge accounting, transaction costs, hard trade and cumulative-position bounds, and validation-based checkpoint selection. For each ticker, results are first averaged over seeds and then aggregated with equal weight across the ten tickers.

4.1 European-Option Panel Results

Table 1 reports the main panel-level comparison. The mean-oriented ProtoHedge frontier achieves the best average mean gains across the ten-ticker panel, while the tail-oriented ProtoHedge frontier achieves the best average empirical $\text{CVaR}_{5\%}$. The result is therefore not that one ProtoHedge configuration dominates every criterion. Rather, the prototype model produces an interpretable mean-tail frontier that is at least as competitive as vanilla Deep Hedging under both objectives.

All models face the same short-call liability, transaction-cost model, and position constraints. Thus, the relative comparison is the central object of interest. The negative level of average gains reflects the conservative constrained evaluation environment and should not be interpreted as evidence that the ordering among hedges is uninformative. Under this common protocol, a less negative gain or CVaR value is better.

Relative to vanilla Deep Hedging, the mean-oriented ProtoHedge frontier has a positive average mean-gain difference of 0.000625 and a positive average $\text{CVaR}_{5\%}$ difference of 0.001030. The tail-oriented frontier has a positive average mean-gain difference of 0.000132

Model	Mean gains	CVaR _{5%}	Shortfall probability	Bound occupancy
Unhedged	-0.030502	-0.140291	0.615941	0.000000
Spot-Delta	-0.030331	-0.137235	0.815792	0.000000
Spot-Delta Band	-0.030295	-0.137247	0.815806	0.000000
Vanilla Deep Hedging	-0.030277	-0.134793	0.979211	0.077515
ProtoHedge (Mean frontier)	-0.029651	-0.133763	0.896937	0.048616
ProtoHedge (Tail frontier)	-0.030145	-0.132121	0.918602	0.030434

Table 1: Average European-option results across the ten-ticker panel. Higher mean gains and higher CVaR_{5%} values are better; lower bound occupancy is better. Shortfall probability is reported as a secondary distributional diagnostic and should be interpreted jointly with the mean and tail-loss measures.

and a larger average CVaR_{5%} improvement of 0.002672. The mean-oriented frontier exceeds vanilla on mean gains in all ten tickers and on CVaR_{5%} in seven tickers. The tail-oriented frontier exceeds vanilla on mean gains in six tickers and improves CVaR_{5%} in all ten tickers.

Table 2 gives the ticker-level comparison underlying these aggregate statements. The mean frontier is reported beside vanilla for average gains, while the tail frontier is reported beside vanilla for CVaR_{5%} and bound occupancy. This presentation makes the intended tradeoff explicit: ProtoHedge is not being evaluated only on a pooled average, but across broad-market, sector, fixed-income, and single-stock underlyings.

Figure 1 visualizes the mean-gain results by ticker and model. Each row should be read horizontally: the cell values are seed-averaged held-out gains for one ticker, and less negative values indicate better performance. The figure shows that the mean-oriented ProtoHedge frontier tracks or improves on vanilla Deep Hedging across the panel, with the clearest gains on AAPL, NVDA, TLT, and XLK.

Figure 2 gives the corresponding tail-risk comparison. It shows that a ProtoHedge frontier is the strongest CVaR_{5%} model for every ticker in the panel. In particular, the tail-oriented configuration improves on vanilla in all ten direct comparisons, which is the strongest panel-level evidence that prototype structure need not come at the cost of tail behavior.

4.2 Position-Bound Analysis

Position bounds prevent a learned hedge from taking unrealistically large cumulative exposures. Bound occupancy therefore measures how often a policy is effectively pressed against the edge of the feasible trading region. A policy with strong performance but persistent bound-saturation is less economically credible, because its apparent performance may depend heavily on the imposed constraint rather than on a stable hedge rule.

The aggregate evidence favors ProtoHedge. Vanilla Deep Hedging has mean bound occupancy of 0.0775, compared with 0.0486 for the mean-oriented ProtoHedge frontier and 0.0304 for the tail-oriented frontier. These correspond to average reductions of approximately 37% and 61%, respectively, relative to vanilla. The improvement is not universal: for TLT, XLE, XLF, and XLV, the selected tail frontier has higher occupancy than vanilla. This heterogeneity is important and shows why the paper reports per-ticker diagnostics rather than only a pooled average.

Figure 3 makes this diagnostic transparent. The learned-model columns should be compared within each row; darker or larger values correspond to more frequent contact with a hard cumulative-position limit. ProtoHedge often achieves competitive mean or tail performance without greater bound reliance, particularly for AAPL, IWM, NVDA, QQQ, and SPY.

The bound analysis strengthens the main performance result. ProtoHedge does not obtain its favorable panel-level mean and CVaR performance by systematically taking more extreme feasible positions. Instead, the prototype bottleneck is consistent with a less boundary-driven policy on average, while preserving competitive hedging outcomes.

4.3 Interpretability Verification

Performance alone does not establish that ProtoHedge is interpretable. We therefore inspect the learned prototype mechanism directly using a representative QQQ frontier model with 25 prototypes. The goal is not to claim that a small number of states completely explains every decision, but to test whether the policy actually routes decisions through a sparse and inspectable set of recurring empirical states.

Figure 4 reports cumulative prototype usage after sorting prototypes from most to least used. The three most-used prototypes account for 58.4% of total assignment mass, and the ten most-used prototypes account for approximately 99.6%. Thus, although the model is supplied with 25 prototypes, its decisions are effectively concentrated in a much smaller subset of recurrently used state representatives.

Figure 5 tests whether this usage is state-dependent rather than static. Each row is a volatility regime and each column is a prototype; color denotes the average similarity weight assigned in that regime. Several dominant prototypes change substantially across regimes. For example, prototype 8 is most prominent in low-volatility episodes, whereas prototypes 18 and 23 receive substantially more weight in high-volatility episodes. This behavior is consistent with the intended interpretation of ProtoHedge: distinct representative states become relevant under different market conditions.

Finally, Figure 6 grounds the dominant prototypes in actual historical episodes. Each panel shows the normalized spot path nearest to one of the most-used prototypes, and the dashed vertical line marks the time step at which the prototype match occurs. The retrieved examples are recognizable market configurations—including moderately rising, recovering, and sharply drawn-down

Ticker	Mean gains		CVaR _{5%}		Bound occupancy	
	Vanilla	Proto Mean	Vanilla	Proto Tail	Vanilla	Proto Tail
AAPL	-0.03963	-0.03787	-0.14793	-0.14637	0.12504	0.00959
IWM	-0.02711	-0.02699	-0.12398	-0.12251	0.09070	0.00275
NVDA	-0.07238	-0.06988	-0.32653	-0.32036	0.17440	0.01160
QQQ	-0.02881	-0.02863	-0.11040	-0.10687	0.16596	0.03044
SPY	-0.02168	-0.02162	-0.08590	-0.08397	0.19267	0.00015
TLT	-0.01430	-0.01339	-0.08194	-0.08115	0.00000	0.08351
XLE	-0.02856	-0.02851	-0.17052	-0.16881	0.00000	0.01507
XLFX	-0.02381	-0.02367	-0.10478	-0.10208	0.00000	0.13615
XLK	-0.02809	-0.02773	-0.10933	-0.10643	0.02638	0.01186
XLV	-0.01842	-0.01822	-0.08661	-0.08266	0.00000	0.00323

Table 2: Ticker-level European-option comparison with vanilla Deep Hedging. Higher mean gains and CVaR_{5%} are better, while lower bound occupancy is better. The two ProtoHedge columns use the screened mean and tail frontiers, respectively.

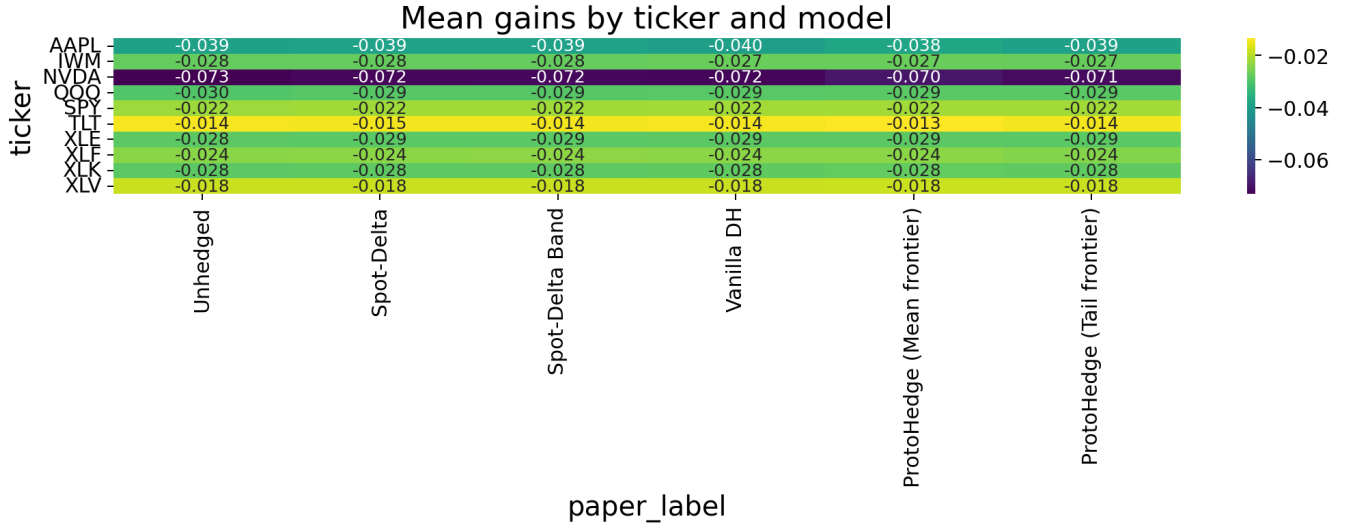


Figure 1: Mean gains by ticker and model for European short-call hedging. Each row is a ticker and each cell is the mean held-out gain averaged over seeds. Comparisons should be made horizontally within a ticker: higher, or less negative, values are better.

paths—rather than abstract latent coordinates. This retrieval property is the practical interpretability advantage of ProtoHedge: a hedge decision can be linked back to concrete historical situations that a domain expert can inspect.

Together, the concentration, regime-usage, and nearest-episode diagnostics provide direct evidence that the prototype layer is active and inspectable in the real-data setting. They do not prove that every prototype has a unique economic interpretation, but they rule out the simpler concern that prototypes are merely decorative parameters with diffuse or state-invariant usage.

4.4 Asian-Option Extension and Completion Criteria

The Asian extension tests the same empirical claim under a more demanding liability. Whereas the European payoff is determined by terminal spot, an arithmetic Asian short call depends on the

entire realized spot path. The European result therefore does not automatically establish robustness to path dependence.

The completed Asian panel will retain the same ten tickers, rolling 20-step episodes, transaction costs, bounds, model grid, train-validation-test splits, and three-seed protocol. The sole deliberate change is the addition of the causal running-average moneyness from Section 3.2 to the state of both learned policies. The resulting comparison will report the same panel-average and ticker-level mean-gain, CVaR_{5%}, shortfall, and bound-occupancy tables as the European experiment. It will also repeat the concentration, regime-usage, and nearest-historical-episode diagnostics for a representative Asian ProtoHedge model.

We do not treat the current preliminary Asian run as a paper-level result. It lacks the running-average state required for a fully observed path-dependent control problem, AAPL did not complete neural training, and QQQ contains only two of the planned three

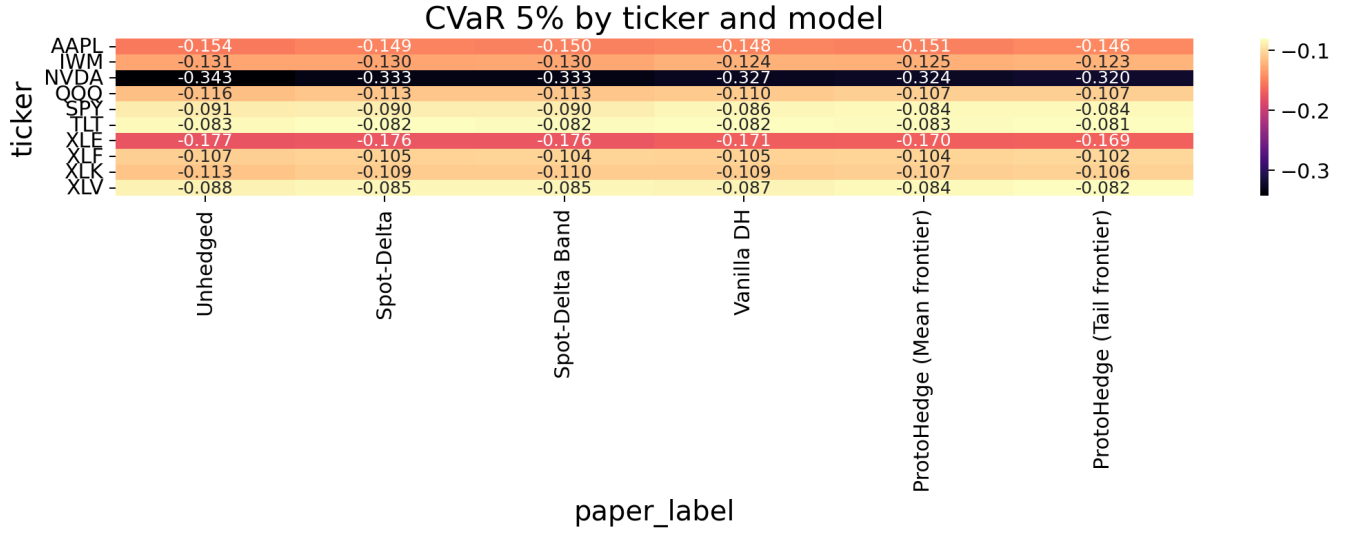


Figure 2: Empirical CVaR_{5%} by ticker and model for European short-call hedging. Less negative CVaR_{5%} values indicate smaller average losses in the worst five percent of held-out episodes. A ProtoHedge frontier provides the strongest CVaR_{5%} result for every ticker.

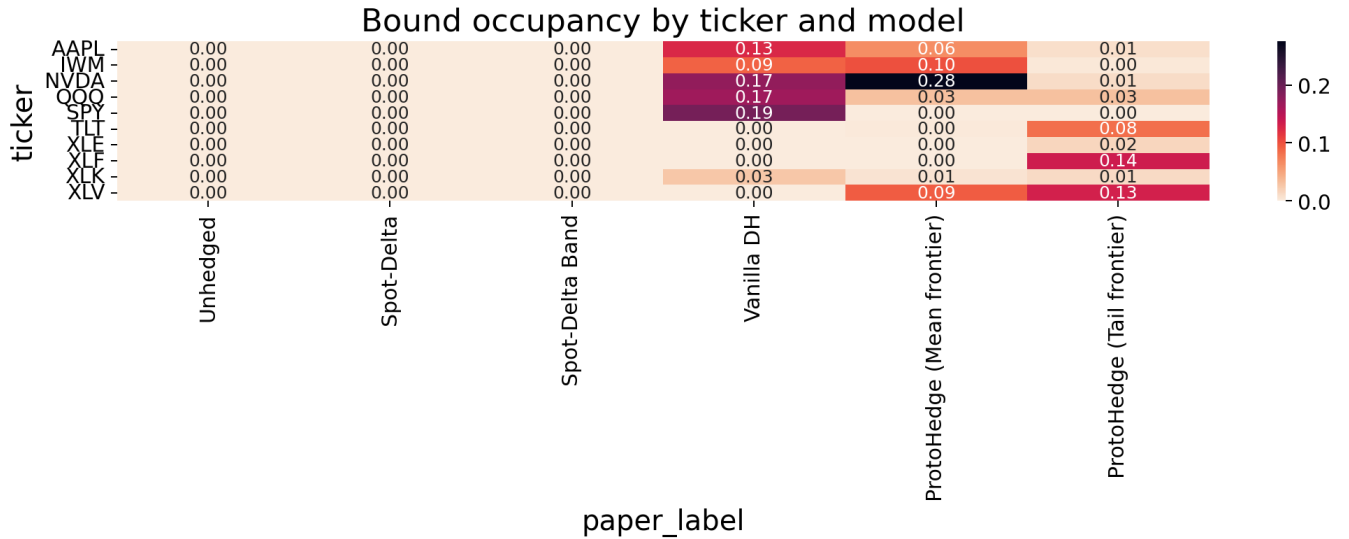


Figure 3: Cumulative position-bound occupancy by ticker and model. Each cell is the fraction of held-out decision times at which the cumulative hedge is at a hard position limit. Lower values are more economically credible. The comparison among learned policies shows that ProtoHedge often reduces reliance on the hard bounds relative to vanilla Deep Hedging.

seeds. These are implementation completion issues, not a finding about ProtoHedge. Reporting the incomplete sweep beside the completed European panel would bias the comparison and overstate the scope of the empirical evidence.

5 DISCUSSION AND LIMITATIONS

The evidence in this paper supports the empirical-validation motivation of ProtoHedge. On a ten-ticker panel of historical

European-option hedging episodes, ProtoHedge remains competitive with standard Deep Hedging while retaining a directly inspectable prototype layer. The mean-oriented ProtoHedge frontier achieves the strongest average mean gains across the panel, while the tail-oriented frontier achieves the strongest average empirical CVaR_{5%}. Both ProtoHedge frontiers also have lower average position-bound occupancy than standard Deep Hedging.

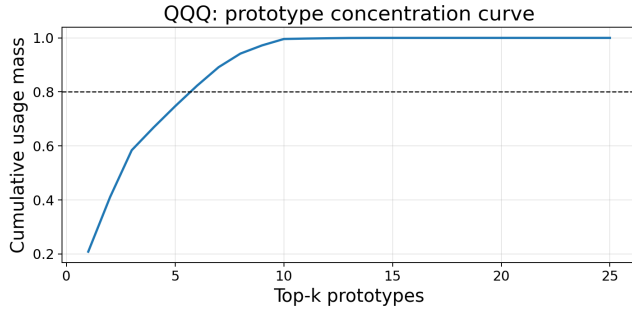


Figure 4: Prototype concentration for a representative QQQ ProtoHedge model with 25 prototypes. The horizontal axis gives the number of most-used prototypes retained, and the vertical axis gives their cumulative assignment mass. The steep initial increase shows that a small subset of prototypes explains most decisions.

The appropriate interpretation is not that ProtoHedge universally dominates all hedging methods. Rather, the results show that imposing a prototype bottleneck need not cause a large empirical performance loss. In the present setting, the bottleneck can coincide with slightly better average gains or tail outcomes, while making the learned policy more auditable. This is the central performance-interpretability tradeoff that ProtoHedge was designed to deliver.

5.1 Where ProtoHedge Works

ProtoHedge performs particularly well on the criteria most closely aligned with the paper’s motivation. First, the mean-oriented frontier improves on standard Deep Hedging in average mean gains for all ten tickers. Second, the tail-oriented frontier improves empirical CVaR_{5%} relative to standard Deep Hedging for all ten tickers. These results span broad-market ETFs, sector ETFs, a Treasury ETF, and large-cap individual equities, suggesting that the finding is not tied to a single underlying.

The position-bound analysis provides a second positive result. On average, both ProtoHedge frontiers spend less time at hard cumulative position limits than standard Deep Hedging. This matters because a policy that repeatedly sits at a constraint may derive apparent performance from extreme feasible exposures rather than from a stable hedging rule. The lower average bound occupancy of ProtoHedge therefore strengthens, rather than weakens, the interpretation of its competitive performance.

Finally, the prototype diagnostics show that the interpretability mechanism remains active on real data. In the representative QQQ analysis, most prototype usage mass is concentrated in a small subset of the available prototypes, the dominant prototypes shift across return, volatility, and drawdown regimes, and prototype matches can be retrieved as concrete historical price paths. This does not mean that every prototype has a single universal economic label. It does mean that the policy can be examined through recurring market states instead of only through opaque neural-network weights.

5.2 Where ProtoHedge Does Not Dominate

The results do not support a claim of universal superiority. The relative advantage varies by ticker, metric, and selected frontier. For example, the tail-oriented ProtoHedge frontier has higher bound occupancy than standard Deep Hedging for several tickers, including TLT, XLE, XLF, and XLV. Similarly, the simple unhedged and delta-style baselines remain competitive on some secondary metrics, particularly shortfall probability under the current gain definition.

This variation is economically plausible. Hedging performance depends on the underlying’s return distribution, implied-volatility behavior, liquidity, option-market quality, and the interaction between transaction costs and position constraints. ProtoHedge should therefore be understood as an interpretable alternative with a favorable average tradeoff, not as a replacement for every black-box or classical hedge in every market.

The current results also do not establish that a single fixed ProtoHedge configuration is optimal across all assets. The reported mean and tail frontiers summarize screened configurations within each ticker. A practitioner seeking to deploy one model specification across markets would need an additional validation exercise that fixes the architecture and hyperparameters before the final test evaluation.

5.3 Limitations

Several limitations define the scope of the present contribution.

First, the study uses a historical U.S. equity and ETF option panel. Although the ten-ticker design is materially broader than a single-underlying study, it does not establish robustness across international equities, equity indices, commodities, foreign exchange, rates, or cryptocurrency markets. The results should therefore not be read as evidence that ProtoHedge is already asset-class invariant.

Second, the hedging environment is realistic in important respects—it uses historical option observations, transaction costs, step-wise accounting, and hard trading constraints—but it remains a controlled backtest. It does not fully model bid-ask execution uncertainty, market impact, funding costs, changing liquidity, margin requirements, or the operational constraints of a live trading desk. The short-call liability is also stylized rather than a complete portfolio of actual client positions.

Third, the current rolling 20-step episode design is well suited to controlled comparison, but future work should include strictly time-ordered out-of-sample tests with non-overlapping evaluation periods. Such tests would provide a stronger assessment of robustness to non-stationarity and would better isolate performance during specific stress events.

Fourth, the interpretability analysis is evidence of inspectability, not external validation of usability. Prototype concentration, regime-dependent usage, and nearest historical states demonstrate that the model exposes meaningful internal structure. However, this paper does not include an evaluation by professional hedgers, portfolio managers, or risk officers. Whether practitioners find the retrieved prototype states useful for actual decision support remains an open question.

Fifth, the study compares ProtoHedge with standard Deep Hedging and classical delta-style baselines, but not with the full range of recent deep-hedging architectures. In particular, future work

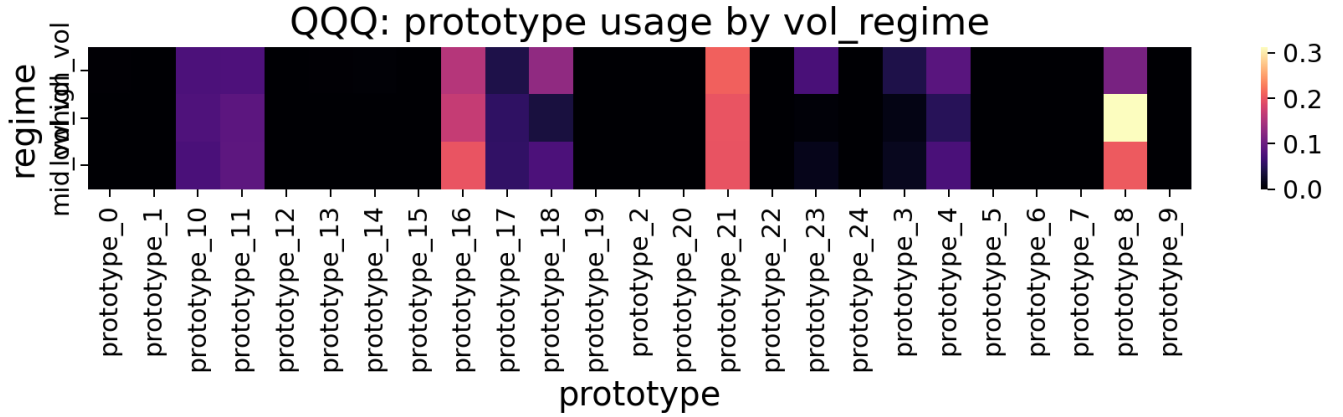


Figure 5: Prototype usage by realized-volatility regime for the representative QQQ model. Rows correspond to low-, medium-, and high-volatility held-out episodes; columns correspond to individual prototypes; color represents average prototype-assignment weight. Regime-specific changes in the dominant columns show that the prototype layer responds to market conditions rather than applying a fixed mixture of prototypes.

should compare against deeper or recurrent hedging networks, alternative risk-aware objectives, and models evaluated under richer dynamics such as Chiarella–Heston-type environments. Those comparisons are necessary to locate ProtoHedge more precisely within the broader deep-hedging literature.

Finally, the present work does not provide a systematic scalability study. We evaluate several prototype counts and similarity specifications, but we do not yet fully characterize how computational cost, prototype count, feature dimensionality, and performance interact as the problem grows. This is especially relevant for high-dimensional hedging tasks involving multiple correlated assets, richer option surfaces, or longer horizons. Approximate nearest-neighbor methods, more efficient prototype-selection procedures, and trainable prototype locations are promising directions for addressing this limitation.

5.4 Path-Dependent and Practical Extensions

The final Asian-option experiment directly addresses the next empirical extension proposed in the original ProtoHedge research agenda. Because an Asian payoff depends on the trajectory of the underlying rather than only its terminal value, it provides a more demanding test of whether prototype-based decisions remain useful under path dependence. Once the full ten-ticker Asian panel is complete, it should be evaluated using the same performance, bound-occupancy, and interpretability framework as the European-option results.

Beyond Asian options, barrier options and other path-dependent liabilities would provide natural stress tests of the framework. Further empirical work should also identify whether dominant prototypes correspond to recognizable market events, including crisis periods, volatility shocks, and rapid drawdowns. The strongest eventual external validation would combine this quantitative analysis with structured feedback from professional hedgers who can assess whether prototype retrievals are economically meaningful and operationally useful.

6 CONCLUSION

This study provides the empirical-validation step that ProtoHedge requires. On a ten-ticker historical European-option panel, prototype-based hedging remains competitive with a matched black-box Deep Hedging benchmark while exposing decisions through a small set of recurring, regime-sensitive, and historically retrievable market states. The mean and tail ProtoHedge frontiers also use less position-bound occupancy on average, supporting the interpretation that their performance does not rely systematically on more extreme feasible exposures.

The path-aware Asian extension remains necessary before generalizing this conclusion to path-dependent liabilities. Its completion requires a causal running-average state and full ten-ticker, three-seed coverage. With that extension in place, the long-form study will provide a stronger foundation from which to prepare a focused conference-length submission.

ETHICS AND PRIVACY STATEMENT

This work uses licensed historical financial-market data and contains no personal data. The models are intended for research on risk management rather than autonomous deployment. Misinterpreting a learned hedge as a production trading recommendation could create financial risk; we therefore report constraints, incomplete experiments, and limitations explicitly.

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A SYNTHETIC REPRODUCTION CHECKS

The repository also reproduces the Black–Scholes and stochastic-volatility comparisons from the original ProtoHedge study. These checks validate the implementation but are not the empirical contribution of this paper.

Model	Black–Scholes	Stochastic volatility
Unhedged	-0.101758	-0.122366
Deep Hedging	-0.059988	-0.089943
ProtoHedge	-0.058237	-0.087943

Table 3: Synthetic expected-utility reproduction checks. Larger values are better.

B ADDITIONAL EUROPEAN DIAGNOSTICS

The paper-assets directory contains the complete per-ticker performance tables, direct ProtoHedge–Deep Hedging gaps, return- and drawdown-regime summaries, and additional prototype-use plots. These diagnostics support the main-text results and permit asset-specific inspection of the panel averages.

QQQ: nearest historical episodes for top prototypes

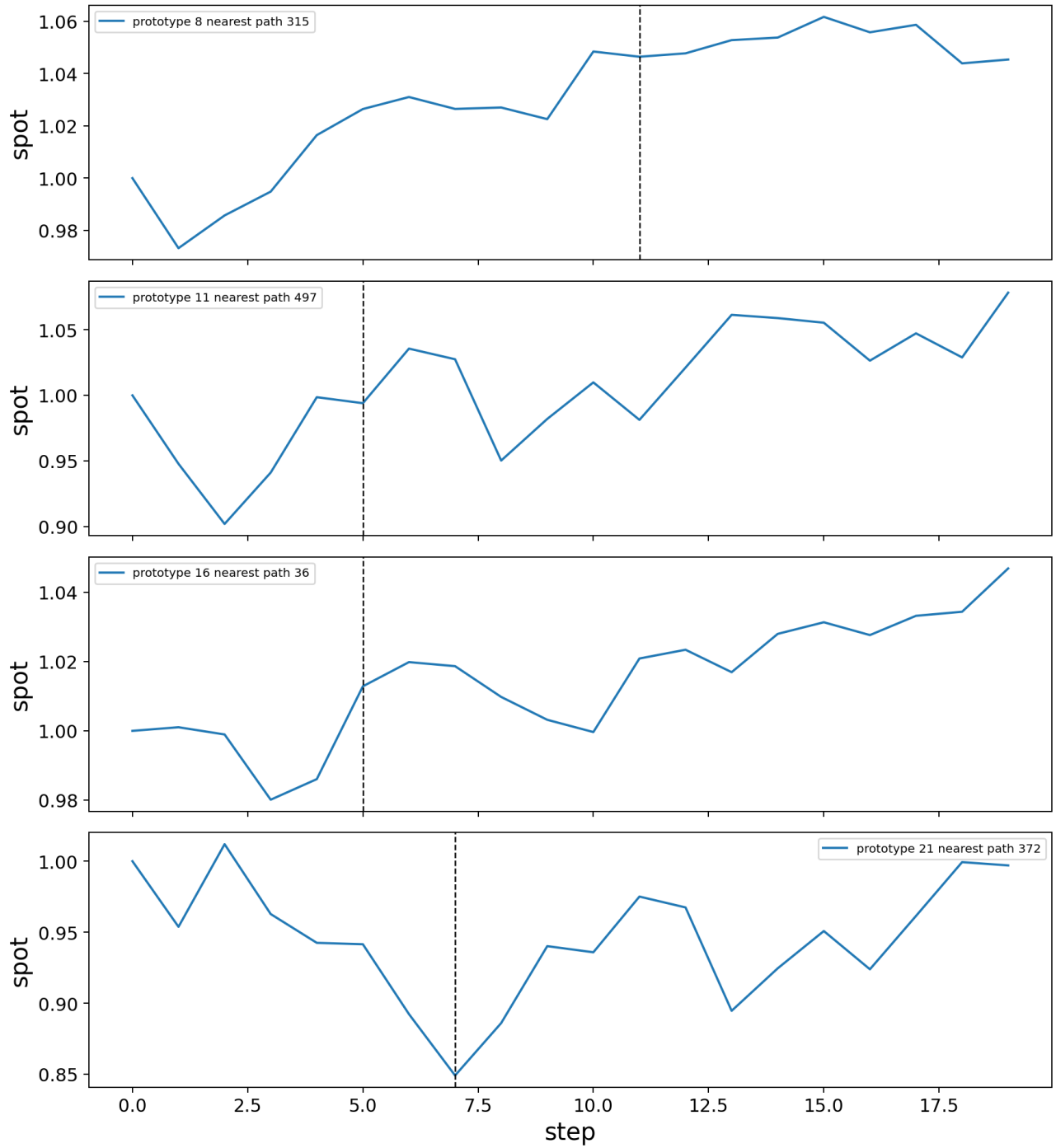


Figure 6: Nearest historical QQQ episodes for four dominant prototypes. Each subplot shows a normalized 20-step spot path; the dashed line marks the state that is nearest to the corresponding prototype. These retrievals connect the learned prototype representation to concrete historical market trajectories.